

Volume Vii Model Theory Reference

Contents

0.1 Languages and Structures	3
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Model Theory

0.1 Languages and Structures

First-Order Languages

Definition 0.1.1 (First-Order Language $\mathcal{L}$). A **first-order language** $\mathcal{L}$ consists of three (possibly empty) sets of non-logical symbols:

- a set $\mathcal{R}$ of **relation symbols** (also called *predicate symbols*), each assigned a fixed arity $n \geq 1$;
- a set $\mathcal{F}$ of **function symbols**, each assigned a fixed **arity** $n \geq 0$ (0-ary function symbols are *constant symbols*);
- a set $\mathcal{C}$ of **constant symbols** (equivalently, 0-ary function symbols — listed separately for emphasis).

Together with the logical symbols common to all first-order languages (variables, connectives $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$, quantifiers $\forall, \exists$, equality $=$, parentheses), the triple $(\mathcal{F}, \mathcal{R}, \mathcal{C})$ determines $\mathcal{L}$.

L-Structures

Definition 0.1.2 ($\mathcal{L}$ -Structure). Let $\mathcal{L} = (\mathcal{F}, \mathcal{R}, \mathcal{C})$ be a first-order language. An **$\mathcal{L}$ -structure** $\mathcal{M}$ consists of:

1. a non-empty set M called the **domain** (or **universe**) of $\mathcal{M}$, written $|\mathcal{M}|$;
2. for each n -ary relation symbol $R \in \mathcal{R}$, a relation $R^{\mathcal{M}} \subseteq M^n$;
3. for each n -ary function symbol $f \in \mathcal{F}$, a function $f^{\mathcal{M}} : M^n \rightarrow M$;
4. for each constant symbol $c \in \mathcal{C}$, a distinguished element $c^{\mathcal{M}} \in M$.

The superscript notation $f^{\mathcal{M}}, R^{\mathcal{M}}, c^{\mathcal{M}}$ denotes the **interpretation** of each symbol in $\mathcal{M}$.

Variable Assignments

Definition 0.1.3 (Variable Assignment). Let $\mathcal{M}$ be an $\mathcal{L}$ -structure with domain M , and let $\text{Var} = \{v_1, v_2, v_3, \dots\}$ be a fixed countable set of first-order variables.

A **variable assignment** (or **environment**) for $\mathcal{M}$ is a function $s : \text{Var} \rightarrow M$.

For a variable assignment s , a variable v_i , and an element $a \in M$, write $s[v_i \mapsto a]$ for the assignment that agrees with s on all variables except v_i , where it returns a :

$$s[v_i \mapsto a](v_j) = \begin{cases} a & \text{if } j = i, \\ s(v_j) & \text{if } j \neq i. \end{cases}$$

Term Evaluation

Definition 0.1.4 (Term Evaluation). Let $\mathcal{M}$ be an $\mathcal{L}$ -structure and $s : \text{Var} \rightarrow M$ a variable assignment. The **evaluation** $s(t) \in M$ of an $\mathcal{L}$ -term t under s is defined inductively:

1. If $t = v_i$ is a variable, then $s(t) = s(v_i)$.
2. If $t = c$ is a constant symbol, then $s(t) = c^{\mathcal{M}}$.
3. If $t = f(t_1, \dots, t_n)$ for an n -ary function symbol f , then $s(t) = f^{\mathcal{M}}(s(t_1), \dots, s(t_n))$.

Satisfaction [Stub]

Elementary Equivalence and Substructures [Stub]

Proofs

Capstone